

TARGETING MEMBRANE STEROL HOMEOSTASIS WITH AZOLES TO INHIBIT PATHOGENIC YEASTS

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Human fungal infections represent a growing threat to global public health and are responsible for approximately 1.6 million deaths each year. Azole antifungals remain among the most widely used treatments. They target the ER-located membrane protein CYP51, a lanosterol demethylase, involved in the ergosterol synthesis pathway. Azole binding on CYP51 disrupts ergosterol biosynthesis and alters the membrane composition, with a reduction of ergosterol content, effectively regulating fungal cell population¹. The main resistance mechanism against azole is the overexpression of the Cdr1 efflux pump that expels azoles from the fungal thus decreasing their intracellular concentration². In this context, we evaluated a new generation of azole compounds designed with two key properties: **(1)** to maintain their antifungal efficacy against CYP51, and **(2)** to inhibit Cdr1-mediated efflux, as a promising strategy to better overcome antifungal resistance. We found the most potent compound, ACH-I-7, with strong broad-spectrum antifungal activity, surpassing that of clinically used azoles. We solved the cryo-EM structure of *Candida glabrata* Cdr1 with ACH-I-7 and with the inhibitor FK506 that deepen our understanding of azole-drug efflux. Collectively, these findings uncovered new avenues for antifungal discovery and established a structure-guided framework for the rational development of improved inhibitors.

REFERENCES

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